Mekala Chandana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.38
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2019	96.70%
St. Arnold's High School	Class X (Board of Secondary Education, Telangana State)	2017	8.80%

SKILLS

Technical Skills: MIKE SHE, HEC-RAS, HEC-HMS, AutoCAD, STAAD. Pro, QGIS, EPANET, MODFLOW

Hydrological Analysis: Flood frequency analysis, Computational discharge measurement, Flood Monitoring.

Statistical Analysis: Spatial data processing & visualization, PCA, Mann-Kendall

Programming Languages: R Programming & Python

PROJECTS

Rainfall Data Analysis of Telangana Districts

- Analyzed 72 years (1950-2021) of rainfall data across four Telangana districts using **RStudio**.

- Developed daily, monthly, seasonal, annual, and decadal analyses; applied Linear Regression, **PCA, Mann-Kendall Test**, and statistical visualization to assess rainfall variability and trends

At-Site Flood Frequency Analysis Using Observed Discharge Data

- Conducted flood frequency analysis using 40 years of observed discharge data from the Manot gauging station.
- Developed skills in hydrological data processing, statistical testing, **goodness-of-fit analysis**, parameter estimation, and flood quantile/return-period estimation.
- Learned about probability distributions (**GEV, Gumbel, Fr´echet**)

Flood Hydrograph simulation and floodplain mapping

- Performance assessment of SAC-SMA, GR4J, and HEC-HMS for flood hydrographs.
- Floodplain mapping for the generated hydrographs using HEC-RAS.

Flood Susceptibility Mapping of the Lower Godavari Basin Using GIS-Based Frequency Ratio Model

- Developed seasonal flood susceptibility maps by integrating DEM, slope, drainage density, flow accumulation, TWI, soil, LULC, rainfall, and historical flood data using **GIS**.
- Applied the **Frequency Ratio (FR) model**, weighted overlay, raster reclassification, and spatial analysis to identify flood-prone zones across the Lower Godavari Basin.

Blended Concrete Using Recycled Fine Aggregate and Sugarcane Bagasse Ash

- Investigated the mechanical and durability properties of concrete incorporating **recycled fine aggregate and sugarcane bagasse ash**, evaluating their potential as sustainable alternatives to conventional concrete materials.

INTERNSHIP EXPERIENCE

AutoCAD Internship | Mini Project - Building Floor Plan & Elevation

- Developed detailed first- and second-floor plans and building elevations for a 1500 sq.ft residential building using AutoCAD.
- Prepared dimensioned architectural drawings, including rooms, doors, windows, staircase, parking, and other building components

Pre-Engineered Building (PEB) Internship | Kirby Building Systems

- Gained practical understanding of **Pre-Engineered Building (PEB) design**, including the underlying design principles and application of pre-engineered components.
- Studied the **advantages and applications of PEB systems** and gained field exposure through a **construction site visit**.

RELEVENT COURSES

- Engineering Hydrology and Hydrologic Systems
- Open Channel Hydraulics and Sediment Transport
- Remote Sensing and GIS
- Advanced Statistical Methods in Hydrology
- Advanced Computational Lab
- Hydraulic and Hydrologic simulation lab
- Computational Fluid Dynamics
- Irrigation Engineering
- R Programming
- Analysis of Water Distribution System
- Groundwater Modelling
- Python